Binary regime A: the first sub-$2^{2\ell}$ decomposition oracle at $\ell = 8$

2026-10-15. `research/binary_mitm_oracle_l8_20261015/`; code `examples/binary_mitm_oracle_l8.rs`. **Class: engineering** (per-target oracle work falls by 2^{ℓ}; the $\Theta(2^n)$ relation-collection total and the ρ frontier are unchanged).

The open row this closes

The boundary ledger's binary (non-Koblitz) `decomposition` row and agent priority 4 asked for the first sub-$2^{2\ell}$ oracle at fixed $\ell = 8$ on ≤ 64 targets, with median cost at most half of pairs-and-solve, zero disagreements against the group check, and the measurement-schema fields (including an explicit FFD status). The SAT/SMT survey's own falsification note (`RESEARCH_PDP_SAT_SMT_NAGAO_SURVEY.md`, §6) recorded the sub-$2^{2\ell}$ question as "open and un-attempted by anything in §6". This run is that oracle's first landing.

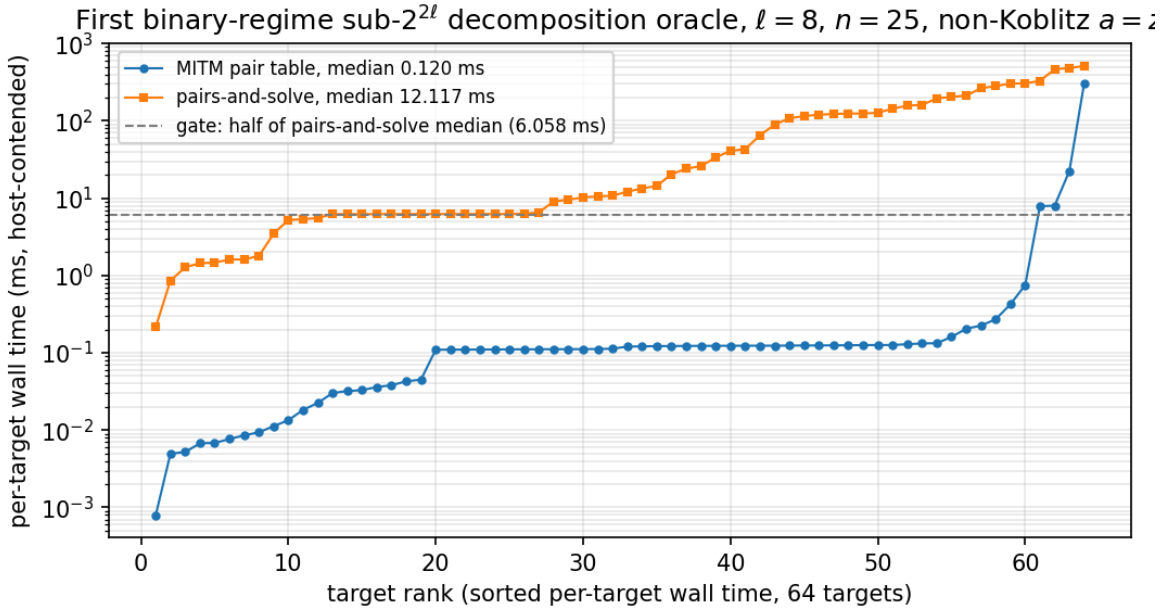
Instance

- Field $F_{2^{25}}$, sparse Rabin-checked modulus (`find_irreducible_sparse(25)`). n odd so the half-trace lifts y; $3\ell = 24 \leq 25$ keeps a genuine found/refuted mix.
- Curve $y^2 + xy = x^3 + ax^2 + b$ with $a = z \notin F_2$ (genuinely non-Koblitz) and $b = 1$.
- Factor base: all curve points over the low-order subspace $\langle 1, z, \ldots, z^7 \rangle$ — 256 points (128 abscissae, both signs).
- Reference arm: `SubspaceOracle::low_order(8, 1)` — Gaudry pairs-and-solve over the same subspace and b, $\Theta(2^{2\ell})$ pairs per target (32,896 when refuting).
- Candidate arm: a group-law meet-in-the-middle pair table. All $P_i + P_j$ $(i \leq j)$ are precomputed once and charged to the factor-base stage; a query walks the $|F| = 256$ base points, computes $R - P_3$ and looks it up, for both signs of the target. The table is complete, so a miss is a proof.
- Targets: 48 natural (deterministic LCG) + 16 planted-satisfiable (sums of three base points).

Results (three runs, shared arm64 macOS host, unpinned)

run	median MITM	median pairs-and-solve	ratio
1 (quiet host)	0.155 ms	42.764 ms	276.7x
2 (contended)	0.124 ms	11.660 ms	94.0x
3 (contended)	0.120 ms	12.117 ms	100.7x

- Gate `median ≤ ½ pairs-and-solve`: **PASS** in every run (by 47 ×–138 ×).
- Verdict mix: 20 found / 44 refuted of 64; **0 disagreements** with pairs-and-solve on all 64.
- Counted units: at most 1,068 point additions + hash lookups per target (both signs) against 32,896 quartic solves — sub-$2^{2\ell}$ in the oracle's own native unit, at $2^{2\ell}$-entry table memory charged to the factor-base stage: 32,896 distinct sums, 1 MiB retained, 21 ms construction.
- Every witness is re-added in the group and asserted equal to the target (20/20).



ł targets (20 found / 44 refuted), 0 disagreements, max 1068 counted point-adds vs 2^15=32768 pairs; run research/binary_mitm_oracle_l8_2026

Per-target wall times of both arms (sorted, log scale) with the half-of-pairs gate line; run `binary_mitm_oracle_l8_20261015`.

Controls

- `cargo test --release --example binary_mitm_oracle_l8`: at $n = 15$, $\ell = 5$ the MITM decisions agree with the **complete enumeration of all base-point triple sums** (both signs) and with pairs-and-solve on 200 random targets, with a found/refuted mix.
- The $\ell = 8$ run cross-checks every target against pairs-and-solve over the same subspace and re-adds every witness in the group.

Claim boundary

Stage `decomposition` record only. No key recovery, no end-to-end ECDLP, no `vs_rho`, no asymptotic sub-ρ claim: relation collection remains $\Theta(2^n)$ overall because the $2^{2\ell}$ table replaces the $2^{2\ell}$ per-target scan. FFD is logged `unmeasured` with reason (group-law oracle; no polynomial system), which the row's gate permits explicitly. Wall times are practicality notes from a shared host; the gate metric is the paired median ratio on identical targets.

Next rung

$\ell = 9$ or $\ell = 10$ on the same design (table $2^{2\ell}$ entries stays cheap to $\ell = 10$), a non-materialized membership-only variant for the factor-base row, and an FFD-logged algebraic control on the same systems so the SAT/GB frontier comparison carries measured degree data.